



COMPILER CONSTRUCTION

Assignment#1

Lexical Analyzer Implementation

Group members

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Automata Designs

Regular Expressions

3.1 Keywords

Regex: (start|finish|loop|condition|declare|output|input|function|return|break|continue|else)

Valid: start, finish, loop, condition, declare, output, input, function, return, break, continue, else

Invalid: Start, LOOP, begin, end, if, while, for

3.2 Identifiers

Regex: [A-Z][a-z0-9_]{0,30}

Valid: Count, Total2, My_var, X, Scanner99, Hello_world

Invalid: count, myVar, 2Count, _name, ALLCAPS, toolongidentifiertthatexceedsthirtyone

3.3 Integer Literals

Regex: [+]?[0-9]+

Valid: 42, +100, -567, 0, 999, -1, +0

Invalid: 12.34, 1,000, 0x1F, --5, 1_000

3.4 Floating-Point Literals

Regex: [+]?[0-9]+\.[0-9]{1,6}([eE][+]?[0-9]+)?

Valid: 3.14, +2.5, -0.123456, 1.5e10, 2.0E-3, +0.1, -9.999999

Invalid: 3., .14, 1.2345678, 3.14e, +.5, -.0

3.5 String Literals

Regex: "([^\\"\\n]|\\\\"\\ntr)*"

Valid: "hello", "say \"hi\"", "line\\break", "tab\\there", "new\\nline", ""

Invalid: "unterminated, "new

line", 'wrong quotes'

3.6 Character Literals

Regex: '([^\n]|\\["\ntr])'

Valid: 'a', 'z', '\', '\\', '\n', '\t', '\r'

Invalid: ", 'ab', 'abc', '\q', unterminated'

3.7 Boolean Literals

Regex: (true|false)

Valid: true, false

Invalid: True, False, TRUE, FALSE, yes, no, 1, 0

3.8 Operators

Arithmetic: (*[+|*|/|-|%]) Valid: +, -, *, /, %, **

Relational: (==|!=|<|=|>|<|>) Valid: ==, !=, <=, >=, <, >

Logical: (&&||!) Valid: &&, ||, !

Assignment: (\+=|-=|*=|/=|) Valid: =, +=, -=, *=, /=

Inc/Dec: (\++|--) Valid: ++, --

Invalid: ^^^, <>, >=, ==, !=, **=, ~~

3.9 Punctuators

Regex: [(){}[\],;:]

Valid: () { } [] , ; :

Invalid: @ # \$ % ^ & ~ ` ?

3.10 Comments

Single-line: ##[^\n]*

Valid: ## this is a comment

##no space needed

123 !@# anything goes

Multi-line: `#\*(([^\*]|\\*+[^*\#])*)\*+##`

Valid: `##` single line `##`

`##` multi

line `##`

`##` nested `##` comment `##` still comment `##`

Invalid: `##` unclosed comment

`*/` wrong syntax `/*`

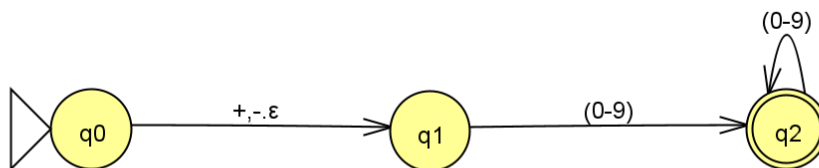
3.11 Whitespace

Regex: `[\t\r\n]+`

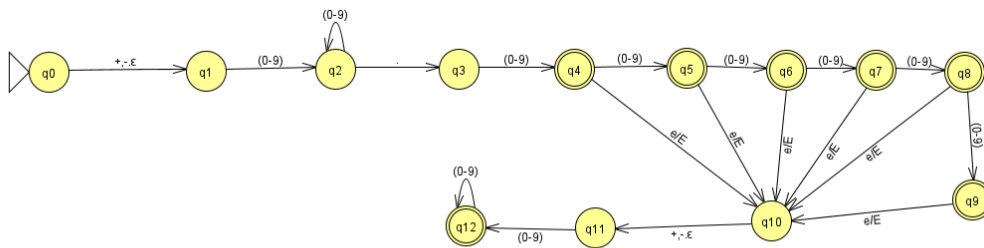
Skipped but line/column tracked: spaces, tabs, carriage returns, newlines

NFA-Diagrams

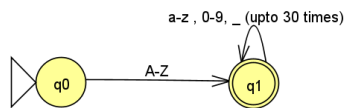
1-Integer Literal



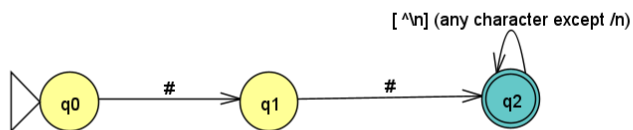
2-Floating Point Literal



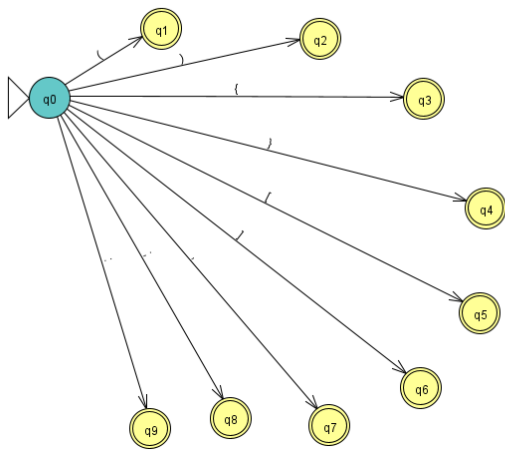
3-Identifier



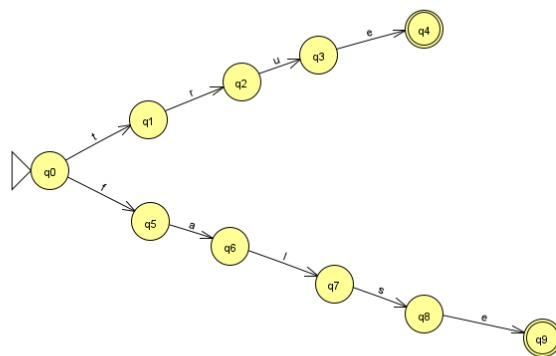
4-Single Line Comment



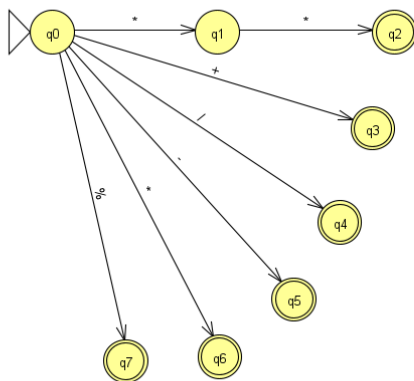
5-Punctuators



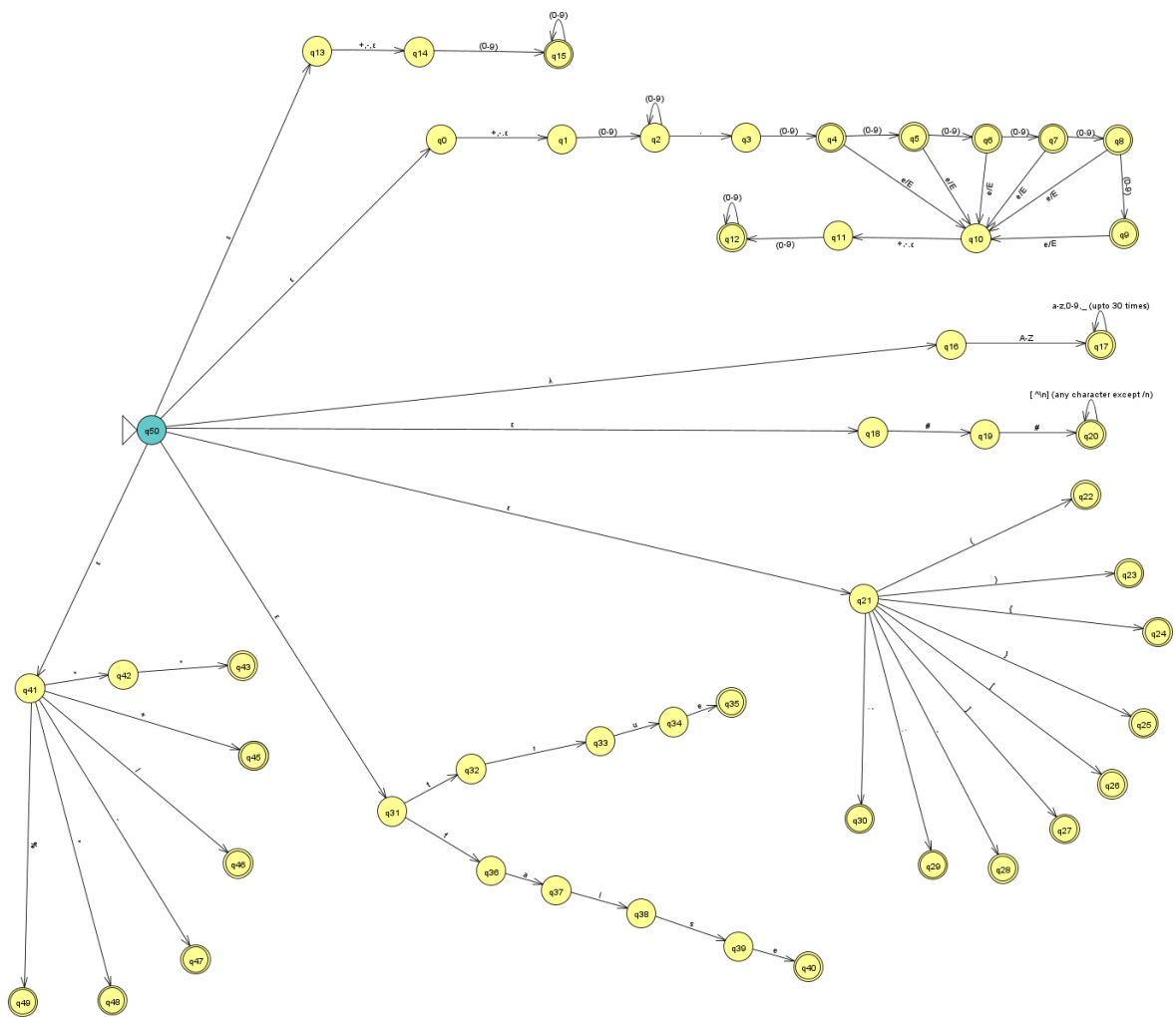
6-Boolean



7-Arithmetic operators



Combined NFA



MINIMIZED DFA TRANSITION TABLE

DFA STATE	A-Z	a-z	0-9	+	-	/	%	.	#	(){};:[]\	Accepting Tokens
Q0	Q1,Q5,Q9	-	Q2	Q17	Q17	Q19	Q20	Q3	Q14	Q16	-
Q1	Q1	Q1	Q1	-	-	-	-	-	-	-	ID
Q2	-	-	Q2	-	-	-	-	Q3	-	-	INT
Q3	-	-	Q4	-	-	-	-	-	-	-	-
Q4	-	-	Q4	-	-	-	-	-	-	-	Float
Q5	Q6	-	-	-	-	-	-	-	-	-	-
Q6	Q7	-	-	-	-	-	-	-	-	-	-
Q7	Q8	-	-	-	-	-	-	-	-	-	-
Q8	-	-	-	-	-	-	-	-	-	-	BOOL
Q9	Q10	-	-	-	-	-	-	-	-	-	-
Q10	Q11	-	-	-	-	-	-	-	-	-	-
Q11	Q12	-	-	-	-	-	-	-	-	-	-
Q12	Q13	-	-	-	-	-	-	-	-	-	-
Q13	-	-	-	-	-	-	-	-	-	-	BOOL
Q14	-	-	-	-	-	-	-	-	Q15	-	-
Q15	Q15	Q15	Q15	Q15	Q15	Q15	Q15	Q15	Q15	Q15	COMMENT
Q16	-	-	-	-	-	-	-	-	-	-	PUNC
Q17	-	-	-	-	-	-	-	-	-	-	ARTH
Q18	-	-	-	Q21	-	-	-	-	-	-	-
Q19	-	-	-	-	-	-	-	-	-	-	ARTH
Q20	-	-	-	-	-	-	-	-	-	-	ARTH
Q21	-	-	-	-	-	-	-	-	-	-	ARTH

MINIMIZED DFA

